

Titration report



Method

Method ID	CDC_M003	Release state	Released
Name	Titration with HCl - Liquid	Modified by	PredefinedUser
Modified on	2/25/2026 4:22:28 PM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.022195 mL
		Concentration R2	0.10044 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.019507 mL
		Concentration R2	0.10039 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.021979 mL
		Concentration R2	0.10044 mol/L

Statistics

R1

Mean	5.021227 mL	Minimum	5.019507 mL
Standard deviation	0.001493 mL	Rel. standard deviation	0.030 %
Total samples	3	Samples excluded	0

R2

Mean	0.10042 mol/L	Minimum	0.10039 mol/L
Standard deviation	0.00003 mol/L	Rel. standard deviation	0.029 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165280	Task name	Unused DG111 - INC=0.1mL, VDISP=2mL
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.022195 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10044 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.019507 mL	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10039 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.021979 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10044 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:24:21 PM
Overall state	OK
Sample start time	2/25/2026 4:24:21 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:24:23 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:24:23 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	2/25/2026 4:24:50 PM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 4:26:51 PM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	100.0 mV
EHNv1	-71.3 mV
Q1	0.502220 mmol
VEQ1	5.022195 mL
EST	-176.8 mV
VEX	0.977805 mL
VEND	6.0000 mL
t	03:05 min
QEND	0.60000 mmol
QEX	0.097780 mmol
EPD	-82.1 mV
nEQ	1

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Consumption (R1)

Creation time	2/25/2026 4:30:01 PM
Overall state	OK
Result	5.022195 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 4:30:01 PM
Overall state	OK
Result	0.10044 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:30:22 PM
Overall state	OK
Sample start time	2/25/2026 4:30:22 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:30:23 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:30:24 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	2/25/2026 4:30:51 PM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 4:32:51 PM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	99.9 mV
EHNv1	-71.6 mV
Q1	0.501951 mmol

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VEQ1	5.019507 mL
EST	-176.9 mV
VEX	0.980493 mL
VEND	6.0000 mL
t	03:03 min
QEND	0.60000 mmol
QEX	0.098049 mmol
EPD	-82.1 mV
nEQ	1

Consumption (R1)

Creation time	2/25/2026 4:35:59 PM
Overall state	OK
Result	5.019507 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 4:35:59 PM
Overall state	OK
Result	0.10039 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 4:36:21 PM
Overall state	OK
Sample start time	2/25/2026 4:36:21 PM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 4:36:22 PM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 4:36:23 PM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:13 min

Stir (Stir1)

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Creation time 2/25/2026 4:36:50 PM
Overall state OK

Titration (EQP) [1] (TitrationEQP1)

Creation time 2/25/2026 4:38:51 PM
Overall state OK
Terminated at Number of EQPs
EEQ1 98.7 mV
EHN1 -72.3 mV
Q1 0.502198 mmol
VEQ1 5.021979 mL
EST -176.4 mV
VEX 0.978021 mL
VEND 6.0000 mL
t 03:02 min
QEND 0.60000 mmol
QEX 0.097802 mmol
EPD -82.5 mV
nEQ 1

Consumption (R1)

Creation time 2/25/2026 4:41:57 PM
Overall state OK
Result 5.021979 mL
Decimal places 6
Substance None
M 1 g/mol
z 1

Concentration (R2)

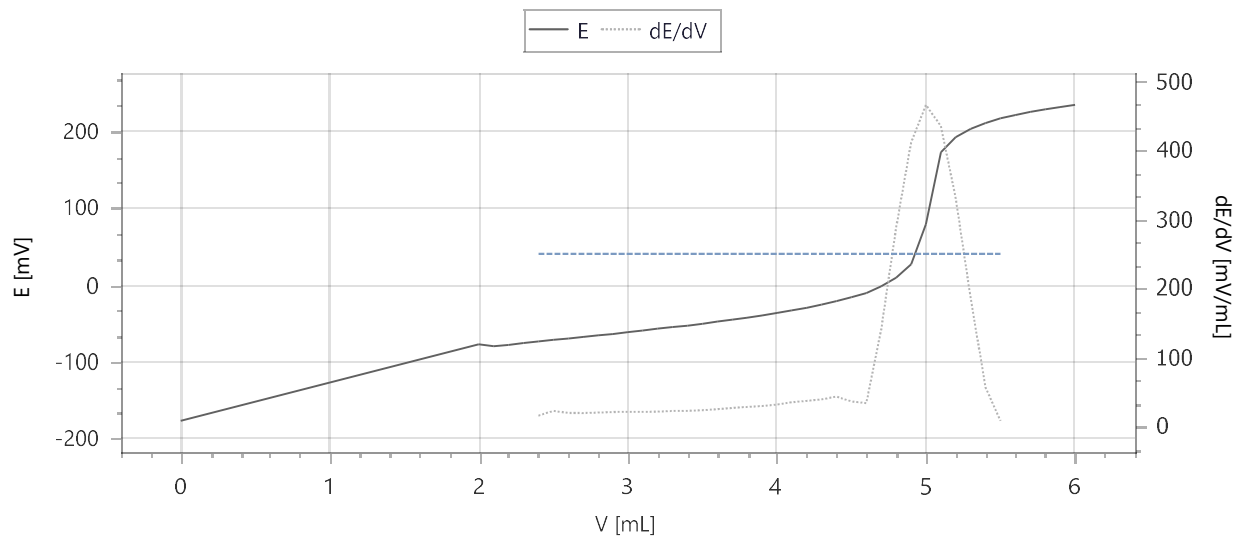
Creation time 2/25/2026 4:41:58 PM
Overall state OK
Result 0.10044 mol/L
Decimal places 5
Substance None
M 1 g/mol
z 1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.8		25.0			
6	2.0000	-77.3	2.0000	25.0	99.5		
40	2.1000	-79.7	0.1000	25.0	-2.4		
43	2.2000	-77.8	0.1000	25.0	1.9		
46	2.3000	-75.6	0.1000	25.0	2.2		
49	2.4000	-73.5	0.1000	25.0	2.1	15.88	
52	2.5000	-71.5	0.1000	25.0	2.0	22.75	
55	2.6000	-69.7	0.1000	25.0	1.8	19.82	
58	2.7000	-67.6	0.1000	25.0	2.1	19.68	
61	2.8000	-65.4	0.1000	25.0	2.2	20.32	
64	2.9000	-63.7	0.1000	25.0	1.7	21.15	
67	3.0000	-61.5	0.1000	25.0	2.2	21.48	
70	3.1000	-59.2	0.1000	25.0	2.3	21.36	
73	3.2000	-56.9	0.1000	25.0	2.3	21.71	
76	3.3000	-54.8	0.1000	25.0	2.1	22.67	
79	3.4000	-52.8	0.1000	25.0	2.0	22.83	
82	3.5000	-50.4	0.1000	25.0	2.4	23.75	
85	3.6000	-47.7	0.1000	25.0	2.7	25.28	
88	3.7000	-45.0	0.1000	25.0	2.7	27.06	
92	3.8000	-42.7	0.1000	25.0	2.3	28.51	
94	3.9000	-39.5	0.1000	25.0	3.2	29.79	
98	4.0000	-36.3	0.1000	25.0	3.2	31.93	
101	4.1000	-33.0	0.1000	25.0	3.3	35.33	
104	4.2000	-29.5	0.1000	25.0	3.5	37.27	
107	4.3000	-25.5	0.1000	25.0	4.0	39.46	
110	4.4000	-20.8	0.1000	25.0	4.7	43.58	
113	4.5000	-15.7	0.1000	25.0	5.1	36.46	
116	4.6000	-10.6	0.1000	25.0	5.1	34.01	
119	4.7000	-1.9	0.1000	25.0	8.7	140.88	
122	4.8000	9.6	0.1000	25.0	11.5	287.85	
127	4.9000	27.2	0.1000	25.0	17.6	410.89	

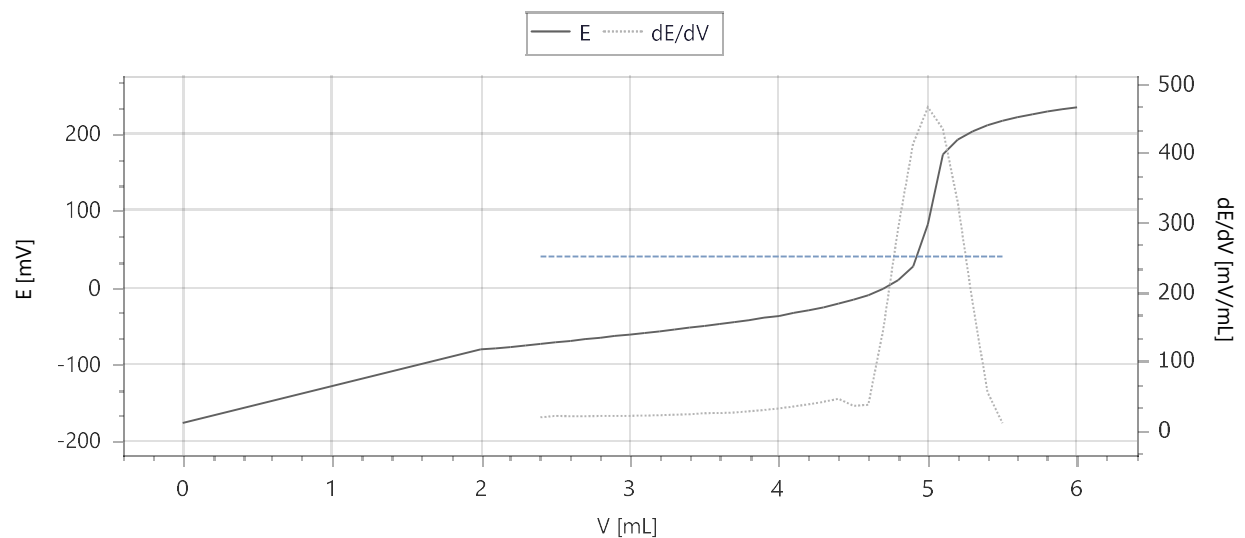
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137	5.0000	79.1	0.1000	25.0	51.9	465.55	EQP
	5.022195	100.0				465.76	
142	5.1000	173.1	0.1000	25.0	94.0	434.99	
145	5.2000	192.4	0.1000	25.0	19.3	330.52	
148	5.3000	203.4	0.1000	25.0	11.0	185.94	
151	5.4000	210.8	0.1000	25.0	7.4	56.73	
154	5.5000	216.9	0.1000	25.0	6.1	8.41	
158	5.6000	221.4	0.1000	25.0	4.5		
161	5.7000	225.6	0.1000	25.0	4.2		
164	5.8000	228.9	0.1000	25.0	3.3		
167	5.9000	231.7	0.1000	25.0	2.8		
170	6.0000	234.5	0.1000	25.0	2.8		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.9		25.0			
6	2.0000	-80.8	2.0000	25.0	96.1		
40	2.1000	-79.7	0.1000	25.0	1.1		
43	2.2000	-77.9	0.1000	25.0	1.8		
46	2.3000	-75.7	0.1000	25.0	2.2		
49	2.4000	-73.7	0.1000	25.0	2.0	18.67	
52	2.5000	-71.8	0.1000	25.0	1.9	20.67	
55	2.6000	-69.9	0.1000	25.0	1.9	20.32	
58	2.7000	-67.7	0.1000	25.0	2.2	20.33	
61	2.8000	-65.7	0.1000	25.0	2.0	20.54	
64	2.9000	-63.5	0.1000	25.0	2.2	20.59	
67	3.0000	-61.6	0.1000	25.0	1.9	20.84	
70	3.1000	-59.5	0.1000	25.0	2.1	21.18	
73	3.2000	-57.4	0.1000	25.0	2.1	21.72	
76	3.3000	-55.1	0.1000	25.0	2.3	22.44	

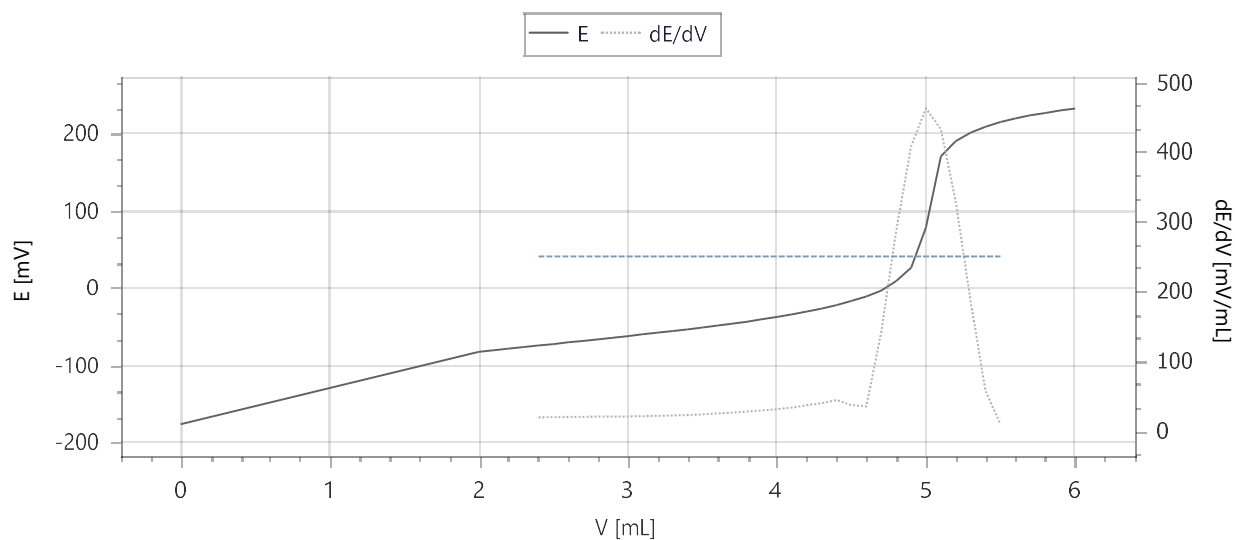
Titration report

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79	3.4000	-52.7	0.1000	25.0	2.4	23.11	
82	3.5000	-50.5	0.1000	25.0	2.2	24.59	
85	3.6000	-48.1	0.1000	25.0	2.4	24.78	
88	3.7000	-45.4	0.1000	25.0	2.7	25.55	
91	3.8000	-42.8	0.1000	25.0	2.6	27.34	
94	3.9000	-39.6	0.1000	25.0	3.2	29.22	
98	4.0000	-37.6	0.1000	25.0	2.0	31.47	
101	4.1000	-33.5	0.1000	25.0	4.1	34.32	
104	4.2000	-29.8	0.1000	25.0	3.7	37.49	
107	4.3000	-26.1	0.1000	25.0	3.7	41.06	
110	4.4000	-21.4	0.1000	25.0	4.7	45.31	
113	4.5000	-16.3	0.1000	25.0	5.1	35.13	
116	4.6000	-10.2	0.1000	25.0	6.1	36.72	
119	4.7000	-2.0	0.1000	25.0	8.2	144.92	
123	4.8000	9.3	0.1000	25.0	11.3	291.00	
128	4.9000	27.0	0.1000	25.0	17.7	411.99	
134	5.0000	82.2	0.1000	25.0	55.2	465.04	
	5.019507	99.9				465.19	EQP
139	5.1000	173.0	0.1000	25.0	90.8	433.64	
142	5.2000	192.2	0.1000	25.0	19.2	327.87	
146	5.3000	203.1	0.1000	25.0	10.9	182.73	
149	5.4000	210.9	0.1000	25.0	7.8	54.71	
152	5.5000	216.8	0.1000	25.0	5.9	10.43	
155	5.6000	221.4	0.1000	25.0	4.6		
158	5.7000	225.3	0.1000	25.0	3.9		
161	5.8000	228.8	0.1000	25.0	3.5		
164	5.9000	232.0	0.1000	25.0	3.2		
167	6.0000	234.6	0.1000	25.0	2.6		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time	V	Meas. value	dV	Temp.	dE	dE/dV	EQP
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[s]	[mL]	[mV]	[mL]	[°C]	[mV]	[mV/mL]	
0	0.0000	-176.4		25.0			
6	2.0000	-82.6	2.0000	25.0	93.8		
40	2.1000	-80.3	0.1000	25.0	2.3		
43	2.2000	-78.3	0.1000	25.0	2.0		
46	2.3000	-76.3	0.1000	25.0	2.0		
49	2.4000	-74.4	0.1000	25.0	1.9	19.71	
52	2.5000	-72.5	0.1000	25.0	1.9	19.94	
55	2.6000	-70.3	0.1000	25.0	2.2	20.24	
58	2.7000	-68.3	0.1000	25.0	2.0	20.53	
61	2.8000	-66.3	0.1000	25.0	2.0	20.77	
64	2.9000	-64.1	0.1000	25.0	2.2	20.90	
67	3.0000	-62.2	0.1000	25.0	1.9	20.91	
70	3.1000	-59.9	0.1000	25.0	2.3	21.46	
73	3.2000	-57.8	0.1000	25.0	2.1	21.77	
76	3.3000	-55.6	0.1000	25.0	2.2	22.43	
79	3.4000	-53.4	0.1000	25.0	2.2	22.91	
82	3.5000	-50.9	0.1000	25.0	2.5	23.92	
85	3.6000	-48.6	0.1000	25.0	2.3	25.34	
88	3.7000	-45.9	0.1000	25.0	2.7	26.43	
92	3.8000	-43.4	0.1000	25.0	2.5	28.03	
94	3.9000	-40.1	0.1000	25.0	3.3	29.59	
98	4.0000	-37.3	0.1000	25.0	2.8	31.40	
101	4.1000	-34.1	0.1000	25.0	3.2	33.76	
104	4.2000	-30.3	0.1000	25.0	3.8	37.19	
107	4.3000	-26.6	0.1000	25.0	3.7	39.85	
110	4.4000	-22.1	0.1000	25.0	4.5	44.56	
113	4.5000	-16.7	0.1000	25.0	5.4	37.27	
116	4.6000	-10.6	0.1000	25.0	6.1	35.36	
119	4.7000	-3.2	0.1000	25.0	7.4	141.49	
122	4.8000	9.0	0.1000	25.0	12.2	286.39	
127	4.9000	26.4	0.1000	25.0	17.4	407.59	
132	5.0000	78.3	0.1000	25.0	51.9	462.17	
	5.021979	98.7				462.38	EQP
138	5.1000	171.1	0.1000	25.0	92.8	432.52	
142	5.2000	190.3	0.1000	25.0	19.2	328.39	
145	5.3000	201.4	0.1000	25.0	11.1	185.77	
148	5.4000	209.1	0.1000	25.0	7.7	57.47	
151	5.5000	215.1	0.1000	25.0	6.0	9.78	
154	5.6000	219.9	0.1000	25.0	4.8		
157	5.7000	223.8	0.1000	25.0	3.9		
160	5.8000	227.1	0.1000	25.0	3.3		
163	5.9000	230.2	0.1000	25.0	3.1		
166	6.0000	232.9	0.1000	25.0	2.7		

Sample Series

Name
Label 1
Label 3
Label 5

Internal ID
Label 2
Label 4
Label 6

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LabX

Comment

Sample

Sample ID 1

Sample ID 2

Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC=0.1mL, VDISP=2mL	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC=0.1mL, VDISP=2mL	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 16:24:19
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	Unused DG111 - INC=0.1mL, VDISP=2mL	25.02.2026 16:24:19

Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Titration report



Responsible User

Time	Login name	Full name
25.02.2026 16:24:19	PredefinedUser	Administrator